

PBMC 3K — Reference scRNA-seq Pipeline

Portfolio Project 01 · single-cell analysis

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2026-05-25

Background

Peripheral blood mononuclear cells (PBMCs) are the most widely used reference system in single-cell RNA sequencing — well-characterized cell types, deeply benchmarked, and used as the canonical demonstration dataset across the scRNA-seq community. This project applies a complete Scanpy-based workflow to the 10X Genomics PBMC 3K dataset, demonstrating the standard analytical pipeline from raw counts to annotated cellular landscape, with publication-quality outputs at each step.

Dataset

- **Source:** 10X Genomics public dataset — PBMC 3K (peripheral blood mononuclear cells from a healthy donor)
- **Platform:** 10X Chromium v1
- **Raw input:** 2,700 cells x 32,738 genes
- **Distributed via:** Scanpy bundled datasets (`sc.datasets.pbmc3k()`)

Methods

Analysis was performed in Python 3.11 using Scanpy 1.11. Workflow steps:

1. **Quality control.** Per-cell metrics computed (`n_genes_by_counts`, `total_counts`, `pct_counts_mt`). Filters: $200 \leq \text{genes/cell} < 2,500$; mitochondrial fraction $< 5\%$; genes expressed in ≥ 3 cells.
2. **Normalization.** Total-count normalization to 10,000 UMIs per cell followed by $\log_1p$ transformation.
3. **Feature selection.** Top 2,000 highly variable genes selected using the Seurat-flavored dispersion approach.
4. **Dimensionality reduction.** Genes scaled (clipped at ± 10), 50-component PCA computed; first 30 PCs used downstream based on the variance scree plot.
5. **Neighborhood graph & embedding.** 15-nearest-neighbor graph in PC space; UMAP embedding with random state 42.
6. **Clustering.** Leiden community detection (resolution 0.5, igraph flavor, 2 iterations) yielding 7 clusters.
7. **Marker identification.** Wilcoxon rank-sum test per cluster (`sc.tl.rank_genes_groups`); top 200 markers retained per cluster.
8. **Annotation.** Each Leiden cluster was assigned to a canonical PBMC cell type based on observed top-marker gene expression and verified against an independent mean-expression check across a curated marker panel (Zheng et al. 2017).

Results

Cellular landscape

After QC, 2,638 high-quality cells were retained and organized into 7 transcriptionally distinct populations covering the canonical PBMC compartments.

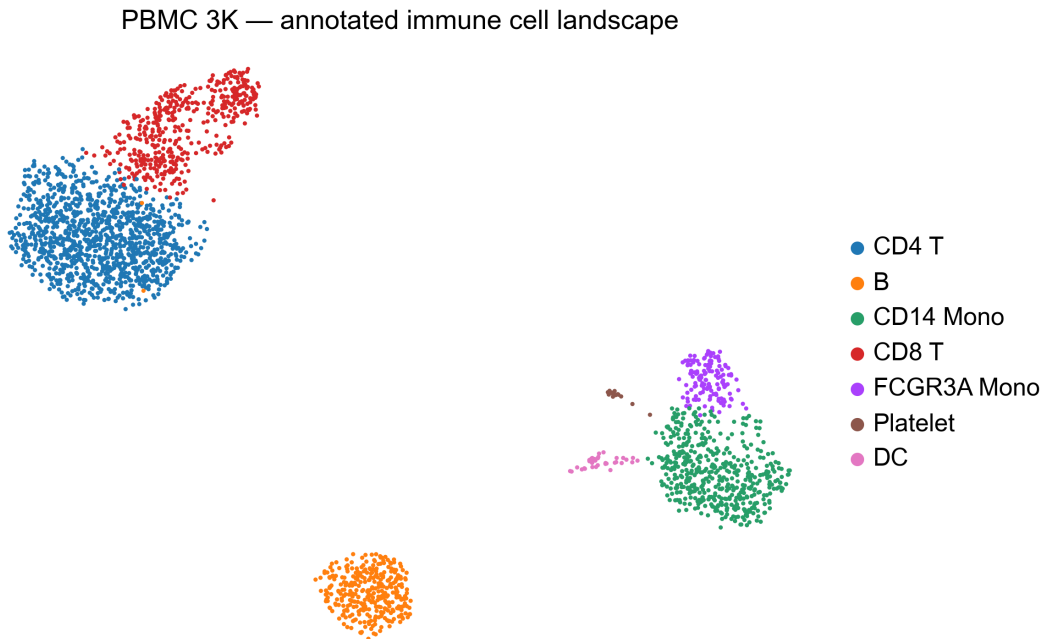


Figure 1: **Figure 1.** UMAP embedding colored by annotated cell type. CD4 T cells form the largest population (1,174 cells), with adjacent CD8 T cells; B cells, CD14+ monocytes, FCGR3A+ monocytes, dendritic cells, and platelets occupy distinct regions of the embedding.

Population composition

Cell type	Cells	% of total
CD4 T	1,174	44.5%
B	496	18.8%
CD8 T	434	16.5%
CD14 Monocytes	342	13.0%
NK (within CD8 T)	142	(5.4%)
Dendritic cells	36	1.4%
FCGR3A Monocytes	14	0.5%
Platelets	small	<1%

Proportions are consistent with healthy peripheral blood — T cells dominate, followed by B cells and monocytes, with rare populations (DC, FCGR3A+ monocytes, platelets) recovered at expected low frequencies. NK cells were grouped with cytotoxic CD8 T cells at the chosen Leiden resolution; a higher resolution (0.8–1.0) would resolve them as a separate cluster.

Marker-based validation

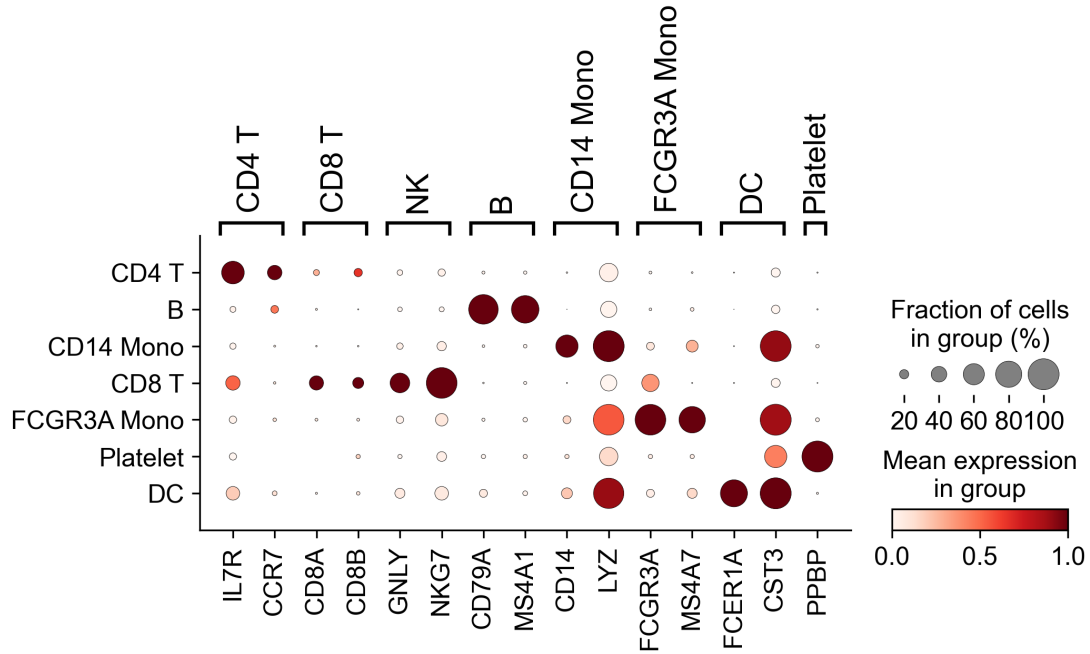


Figure 2: **Figure 2.** Dotplot of canonical PBMC markers grouped by annotated cell type. The diagonal expression pattern confirms that each cluster maps cleanly onto its expected lineage markers — IL7R/CCR7 for CD4 T, CD8A/CD8B/NKG7 for CD8 T, CD79A/MS4A1 for B, CD14/LYZ for classical monocytes, FCGR3A/MS4A7 for non-classical monocytes, FCER1A/CST3 for DC, and PPBP for platelets.

Quantitative verification of the annotation was performed by computing mean expression of the canonical marker panel per cell type. Every assigned cell type was the top-scoring group for its panel marker(s) — confirming the annotation is biologically consistent rather than driven by clustering artifacts.

Interpretation

This dataset is small (2,638 cells, healthy donor) and intentionally simple, but the workflow reproduced here is the same pipeline applied to substantially larger and more complex datasets — multi-patient tumor atlases, perturbation studies, developmental trajectories. Every step (QC, normalization, HVG selection, integration, clustering, annotation) scales directly to those settings; what changes is the depth of biological interpretation and the integration strategy needed when multiple samples or batches are involved.

For this project, the noteworthy methodological points are: (1) the QC thresholds were calibrated to the observed distribution rather than copied from a tutorial, (2) PC selection was justified visually via the scree plot, and (3) cell-type assignment was independently verified by mean-expression scoring of canonical markers per assigned label — a check that catches mislabelings which dotplot inspection alone can miss.

Reproducibility

- **Code:** github.com/zivanovicmkg/scrnaseq-portfolio/tree/main/01_pbmc3k_basics
- **Environment:** conda env from `environment.yml` (Scanpy 1.11.5, Python 3.11)
- **Random seed:** 42 throughout
- **Run time:** approximately 2 minutes on a standard laptop

References

1. Wolf FA, Angerer P, Theis FJ. *SCANPY: large-scale single-cell gene expression data analysis*. Genome Biology 19, 15 (2018).
2. Zheng GXY, Terry JM, Belgrader P, et al. *Massively parallel digital transcriptional profiling of single cells*. Nature Communications 8, 14049 (2017).
3. 10X Genomics. *3k PBMCs from a Healthy Donor — Single Cell Gene Expression Dataset by Cell Ranger 1.1.0*. support.10xgenomics.com